

Modelling the atmosphere



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Abstract

Some simple models for describing the surface temperature of the Earth are introduced. The first of these models neglects the atmosphere's role and thus underestimates the temperature at the surface of the Earth. We then show how the Earth's surface temperature is increased when a simple model for the atmosphere is introduced. This model is also shown to predict a surface temperature for the Earth that is higher than the air temperature, which agrees with the experimental results. However, several factors that affect these temperatures are clearly missing from our model as it overestimates both the surface and air temperatures.

1 Introduction

Environmental scientists have christened the current epoch, the Anthropocene, to reflect that human activity significantly impacts the Earth's geology, landscape, limnology, ecosystems and climate. Much of our impact on these life-supporting systems has been unintentional. At most points in human history, we did not understand our environment well enough to understand the effects we were having. Recently, however, we have developed sophisticated mathematical models that can describe how our collective behaviours affect the planet. Further development of these tools is essential as it will allow us to understand these systems better and inform us of the changes in behaviour that will be necessary to prevent catastrophic change.

This report focuses on models that describe the Earth's climate and surface temperature. To understand the temperature of the Earth, we need to understand the physical processes by which energy is transferred in and out of the climate system. Energy is transferred into the Earth's climate system as it absorbs shortwave radiation from the sun. Meanwhile, energy is transferred out of the climate because the Earth emits radiation into space as it acts like a blackbody emitter. A model for describing these two processes will be described in the following sections. We will show that this model does not describe the Earth's surface temperature well. We will thus refine this basic model by introducing a new model that offers a simple description of the atmosphere. This refined model is a little better because the results we obtain from it reproduce more of the experimental data about the temperature of the Earth. However, we find that there are still significant discrepancies between our refined model and

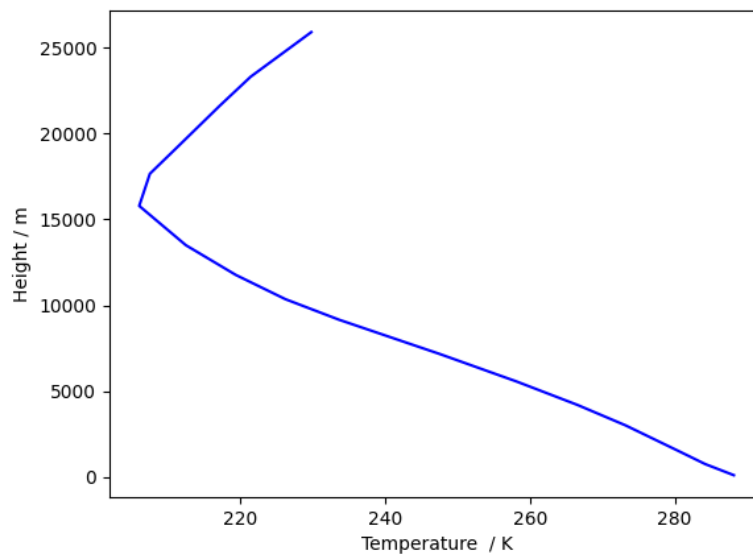


Fig. 1. Annual averaged observed air temperature profile for the atmosphere obtained from the NCEP reanalysis data set.

experimental data. Towards the end of our report, we thus conclude that that our model does not provide a complete description of all the processes via which energy is transferred into and out of the atmospheric system and reflect on its limitations.

The remainder of this report is laid out as follows. In section 2, we first describe the experimental data on the temperature in the climate. Section 3 then describes a model for the temperature that neglects the effect the atmosphere has and shows that it is unable to reproduce the experimental data. Section 4 then introduces the refined model that includes the effect of the atmosphere. The report finishes with conclusions in section 5.

2 Experimental data

Figure 1 shows how the atmosphere's temperature changes as you move away from the surface of the Earth. This graph was created using global, annual averaged observed air temperature from the NCEP reanalysis data, which was copied from [1]. To convert the pressures in this data set to heights above the Earth's surface, we used the calculator at [2].

You can see that the temperature at the surface of the Earth is about 288 K or 15°C. The temperature of the atmosphere decreases until you reach a height of approximately 15 km. After this point, the temperature starts to increase.

3 A basic climate model

As discussed in the introduction, energy from the sun is transferred into the climate system as shortwave radiation. The absorbed solar radiation (ASR) is given by,

$$\text{ASR} = (1 - \alpha)Q$$

where Q is the area-averaged incoming solar radiation, which has been measured as 341.3 W m^{-2} . α , meanwhile, is the planetary albedo and is a (unit-less) measure of the fraction of the radiation that the Earth reflects. Measurements suggest that this $\alpha = 0.29$.

Any planet with a finite surface temperature, T_s , will act as a black body and thus emit long-wave radiation into space. The outgoing, longwave radiation (OLR) can be calculated using:

$$\text{OLR} = \sigma T_s^4,$$

where $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ is the Stefan-Boltzmann constant.

If the Earth is in equilibrium, then the net flow of energy in/out of it should be zero. Consequently, if the only processes that transfer energy in and out of the Earth are ASR and OLR, we should have:

$$\text{ASR} = \text{OLR} \quad \rightarrow \quad (1 - \alpha)Q = \sigma T_s^4$$

By rearranging this expression and inserting the values above, we can thus arrive at the following value for the surface temperature:

$$T_s = \sqrt[4]{\frac{(1 - \alpha)Q}{\sigma}} = \sqrt[4]{\frac{(1 - 0.29)341.3}{5.67 \times 10^{-8}}} = 255.68 \text{ K}$$

You can see that this temperature is approximately 33 K colder than the actual surface temperature of the Earth in figure 1. Furthermore, in this model, the atmosphere's temperature does not depend on the height above the surface of the Earth.

4 Incorporating the atmosphere

We can refine the model introduced in the previous section by describing the Earth as a two-layer system with a surface and an atmosphere. In what follows, we will assume that the atmosphere is transparent to the incoming solar radiation. However, we will also assume that it is opaque to the long-wave radiation emitted by the surface. The only way of transferring energy out of the Earth system is for the atmosphere to behave like a black body and radiate heat outwards. Notice, however, that, as shown in figure 2, black body radiation will also be transferred from the atmosphere to the surface via a process known as back radiation.

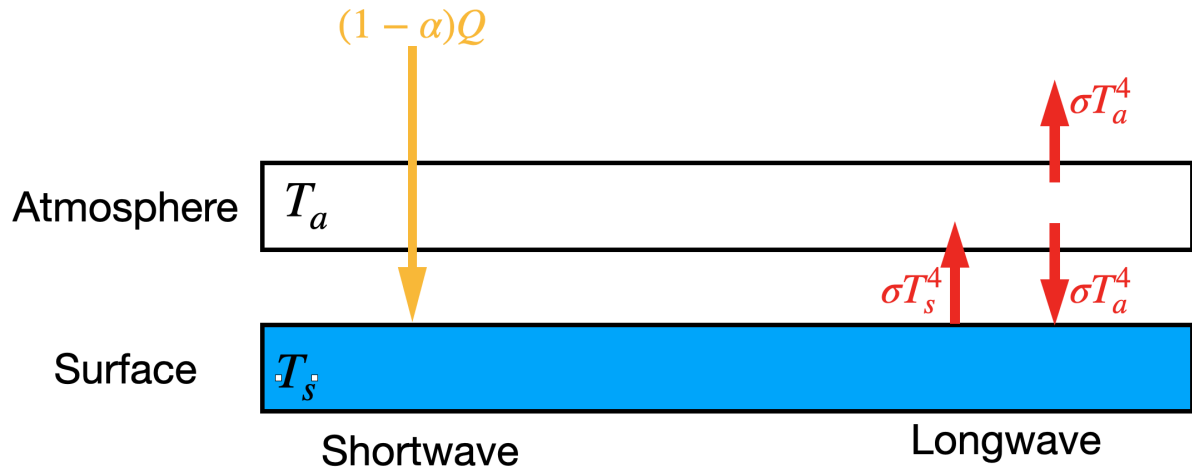


Fig. 2. Schematic showing the energy flows within the two-layer atmosphere model we have used in this report. In this model, energy flows between the environment and the two levels that represent the atmosphere and the Earth's surface. These two levels are at temperatures of T_a and T_s , respectively. The direction of the arrows indicates the layer that the energy flows out of and the layer it flows into. Arrows that do not start/end in one of the two layers represent energy flowing between the system and the environment.

As with the one-layer system described in the previous section, there must be no net transfer of energy in, out or between the two layers of this system. For the surface layer, we must, therefore, have:

$$(1 - \alpha)Q + \sigma T_a^4 = \sigma T_s^4$$

while for the atmosphere, we must have:

$$\sigma T_s^4 = 2\sigma T_a^4$$

These two equations can be rewritten as the following system of linear equations:

$$\begin{pmatrix} -1 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} T_s^4 \\ T_a^4 \end{pmatrix} = \begin{pmatrix} -\frac{(1-\alpha)Q}{\sigma} \\ 0 \end{pmatrix}$$

Values for T_s^4 and T_a^4 were obtained by solving this set of linear equations using NumPy [3]. The values for the temperatures shown in red in figure 3 were then obtained by taking the fourth root of these solutions. When constructing figure 3, we arbitrarily decided to place the boundary between the Earth's surface and the atmosphere 1000 m above sea level.

Figure 3 shows that the atmosphere is colder than the Earth's surface for our model, in agreement with what is found in experiments. However, our assumptions when constructing this model are clearly too severe, as our predictions for the surface and air temperature are much greater than the measured values for these quantities.

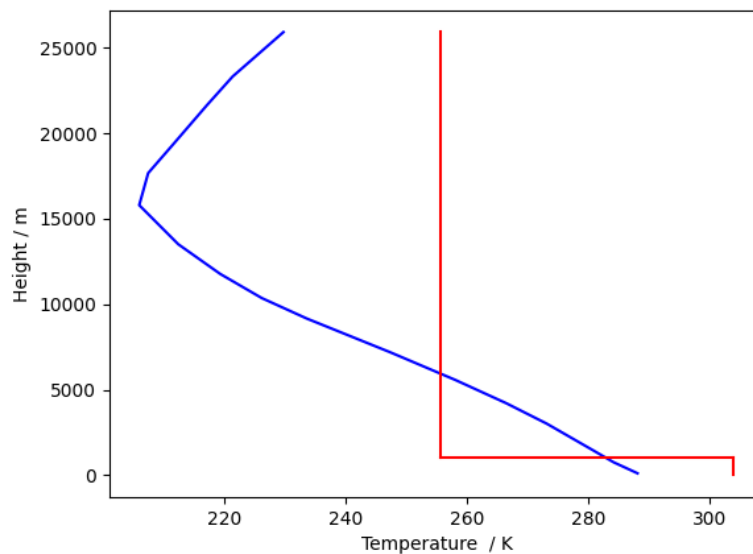


Fig. 3. A comparison between the temperature profile for the two-level model of the atmosphere (red) and the experimental temperature profile in the atmosphere (blue). In drawing the temperature profile for the model, we arbitrarily set the boundary between the Earth's surface and the atmosphere at an altitude of 1000 km.

5 Conclusions

Some simple models that describe how heat is exchanged between the atmosphere, the Earth's surface and the rest of the universe have been introduced. We showed that when the atmosphere is not included in these models, the Earth's surface temperature is underestimated. A simple model that describes the effect of the atmosphere correctly predicts that the air temperature is lower than the surface temperature. However, this refined model overestimates both the air and surface temperature.

The models included here neglect many features that will affect the surface profile in the atmosphere. Some of the assumptions in our model that are particularly severe include

1. The assumption that the atmosphere is opaque to long-wave radiation. The atmosphere will likely only absorb a fraction of the radiation emitted from the Earth's surface rather than all of it.
2. That the atmosphere has no internal structure. In more sophisticated models, the environment is assumed to consist of multiple layers that exchange energy.

Factors such as convection, cloud formation, seasons, ice growth, and ice melting will also significantly affect the Earth's temperature. In short, the models we have introduced in this report only scratch the surface of what is required to successfully model the Earth's climate.

6 References

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